



A Level • Cambridge (CIE) • Chemistry

15 mins 8 questions

Multiple Choice Practice

Topical Practice MCQ

1. Atomic structure

Carefully selected past paper questions by topic.

4 In which pairs are **both** species free radicals?

- 1 Cl and O
- 2 Cl^- and O^{2-}
- 3 Cl and O^-
- 4 Cl^+ and O^{2+}

A 1, 3 and 4 **B** 1 and 3 only **C** 1 only **D** 2 only

- 1 Element X has six more protons than element Y.

Which statement **must** be correct?

- A Atoms of element Y are smaller than atoms of element X.
- B Element X has a full shell of electrons.
- C Element X and element Y are in the same group.
- D Element X and element Y are in the same period.

12 Four successive ionisation energies (IE) of element E are shown.

Element E is in Period 3 of the Periodic Table.

fifth IE / kJ mol^{-1}	sixth IE / kJ mol^{-1}	seventh IE / kJ mol^{-1}	eighth IE / kJ mol^{-1}
16 000	20 000	24 000	29 000

In which group of the Periodic Table is E?

A 14

B 15

C 16

D 17

Question 4

9701_w20_qp_11 • Q3

- 3** A single ^{32}P nucleus can be produced when a single ^{32}S nucleus joins with particle X. In the process a proton is emitted.

What is particle X?

- A** a deuteron, $^2_1\text{H}^+$
- B** an electron
- C** a neutron
- D** a proton

- 3** Which statement about a 3p orbital is correct?
- A** It can hold a maximum of 6 electrons.
 - B** It has the highest energy of the orbitals with principal quantum number 3.
 - C** It is at a higher energy level than a 3s orbital but has the same shape.
 - D** It is occupied by one electron in an isolated phosphorus atom.

10 Element X has a higher first ionisation energy than element Y.

Two students state what they believe is one factor that helps to explain this.

student 1 "X has a higher first ionisation energy than Y because an atom of X has more protons in its nucleus than an atom of Y."

student 2 "X has a higher first ionisation energy than Y because X has a smaller atomic radius than Y."

Only **one** of the two students is correct.

What could X and Y be?

	X	Y
A	carbon	boron
B	magnesium	aluminium
C	oxygen	nitrogen
D	oxygen	sulfur

Question 7

9701_w18_qp_11 • Q1

- 1 The first four ionisation energies for element X are shown in the table.

ionisation energy	1st	2nd	3rd	4th
value / kJ mol^{-1}	577	1980	2960	6190

Which ion of X is produced by removing an electron from a filled shell?

- A** X^+ **B** X^{2+} **C** X^{3+} **D** X^{4+}

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Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_12	1. Atomic structure	A
2	9701_s23_qp_11	1. Atomic structure	B
3	9701_s23_qp_11	1. Atomic structure	A
4	9701_w20_qp_11	1. Atomic structure	C
5	9701_m19_qp_12	1. Atomic structure	D
6	9701_s18_qp_11	1. Atomic structure	D
7	9701_w18_qp_11	1. Atomic structure	D
8	9701_w18_qp_13	1. Atomic structure	D